MSG No. 86.69 $P4'_2/n$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-4'..`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,2,7,8]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[3,4,5,6]`

Table 2: Wyckoff site: `2b`, site symmetry: `-4'..`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,2,7,8]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[3,4,5,6]`

Table 3: Wyckoff site: `4c`, site symmetry: `-1`

No.	position	mapping
1	$[0, 0, 0]$	`[1,3]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,4]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[5,7]`
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[6,8]`

Table 4: Wyckoff site: `4d`, site symmetry: `-1`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,3]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,4]`
3	$\left[0, \frac{1}{2}, 0\right]$	`[5,7]`
4	$\left[\frac{1}{2}, 0, 0\right]$	`[6,8]`

Table 5: Wyckoff site: `4e`, site symmetry: `2..`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,2]`
2	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[3,4]`
3	$\left[\frac{3}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[5,6]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[7,8]`

Table 6: Wyckoff site: `4f`, site symmetry: `2..`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,2]`
2	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[3,4]`
3	$\left[\frac{3}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[5,6]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[7,8]

Table 7: Wyckoff site: 8g, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[2]
3	$[-x,\ -y,\ -z]$	[3]
4	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	[4]
5	$\left[-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[5]
6	$\left[y+\frac{1}{2},\ -x,\ z+\frac{1}{2}\right]$	[6]
7	$\left[y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[7]
8	$\left[\frac{1}{2}-y,\ x,\ \frac{1}{2}-z\right]$	[8]